

Sina Hazeghi

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EDUCATION

Rutgers University - New Brunswick

December 2024

- Bachelor of Science in Computer Science and Mathematics
- GPA: 3.96/4.00 ; SAS Academic Excellence Scholarship Recipient
- Coursework: Artificial Intelligence, Advanced Robotics, Algorithms, Data Science, Data Structures, Statistics, Systems Programming, Software Methodology, Numerical Analysis, Linear Algebra, Stochastic Processes, Computer Architecture

SKILLS

- Languages: Java, Python, C/C++, JavaScript, Typescript, HTML, CSS, X86 Assembly, SQL, R
- Technologies: AWS, CUDA, Git, JAX, Docker, GCP, Pandas, PyTorch, HuggingFace, fast.ai, Transformers, Access DB
- Operating Systems: Linux, MacOS, Unix, Windows, Android, Ubuntu, WSL

WORK EXPERIENCE

Pfizer

May 2024 - August 2024

Process Development Intern

Pearl River, NY

- Deployed Shiny for Python website to automate inventory processes and database integration saving 2-3 hours/shipment
- Updated R-Shiny app and underlying SQL database by merging subroutines to add image compatibility, PDF integration with LaTeX, and dynamically populated protocols, simplifying analyst workflows by 60%
- Manipulated data across multiple databases using Python, Pandas and SQLAlchemy, streamlining and merging over 30 tables into 7, and summarizing data into reports to indicate actionable insights

Rutgers University

December 2023 – PRESENT

Student Researcher robotics.cs.rutgers.edu/pracsys/

New Brunswick, NJ

- Engineered an optimized physics simulator using Data Oriented Design and ECS programming with C++ , prototyped with Python and Taichi to simulate NASA's Tensegrity robot model over 100x faster than MuJoCo
- Optimized Graph Neural Network Inference by 76% using PyTorch on TensorRT engine with dynamic quantization as well as TorchDynamo, allowing for open-loop control for terrain-based research applications

Rutgers University

September 2023 – December 2023

Software Engineer cs.rutgers.edu

New Brunswick, NJ

- Developed a secure Autograder platform using Scapy, Pytest, and Docker to evaluate over 800+ programming projects
- Engineered multithreaded Python scripts for concurrent server-client operations, decreasing grading time by 50%

General Autonomy

July 2023 – October 2023

Software Engineer Intern genauto.ai

Piscataway, NJ

- Spearheaded full-stack software automating 70% of autonomous system safety evaluation using fine-tuned LLMs
- Architected OOP backend in Python with serializable generative graphs, enhancing traceability and cutting memory by 34%
- Prototyped frontend interface serviced with webAPIs from Flask HTTP server and deployed to a custom AWS EC2 instance

PROJECTS

Report Parser Document Processing AI webApp

- Created OCR AI application hosted with GCP to process handwritten police reports into spreadsheets, saving 5+ hours/use
- Utilized Fastai library to clean and augment dataset used for finetuning resnet18 image model for document classification
- Leveraged Google Document-AI to establish data engine to semi-automate labeling and annotating 125+ training documents
- Applied Hugging Face Transformers library to train LayoutLM model to parse handwritten fields into Pandas, which is then cleaned and formatted with Regex and NLP methods to flag inaccurate entries for human review

Maze Solver Motion Generation Implementations

- Devised a custom task in Python and MuJoCo, for a robot to navigate a maze of obstacles from random start points
- Implemented Kinodynamic RRT planner with a 96% success rate, achieving navigation on average of 5.08 seconds
- Devised GPU-accelerated training scripts using Torch RL to leverage distributed training to reduce training time by 95%
- Evaluated A2C, PPO and SAC Reinforcement learning algorithms with various advantage estimation and policy formulation methods from scratch in PyTorch to maximize successful navigation
- Tuned algorithms to surpass stable-baselines3 implementations, achieving a 22% higher success rate after 10,000 training episodes, with successes taking on average only 2.38 seconds